2D Two-Walker Encounter With Shortcut

Robust A1/A8 Verification and Double-Peak Construction

valley-k-small report extension

April 27, 2026

Abstract

This extension delivers two implementation-ready blocks. First, we provide a robust numerical routine to verify the equivalence between appendix Eq. (A1) and Eq. (A8) in the "Product of two generating functions" section, including contour selection, adaptive discretization, and diagnostics. Second, we construct a 2D two-walker encounter model with directed shortcuts and show a shortcut-controlled transition from single peak to double peak in encounter first-passage distributions. All figures/tables/data are generated by one script and are reproducible.

1 Problem Statement

We target two tasks:

1. a numerically stable verification of

$$\frac{1}{2\pi i}\oint_C \frac{du}{u}\,\widetilde{P}_{n_0}(n,u)\widetilde{P}_{m_0}\Big(n,\frac{z}{u}\Big) = \sum_{k_1,k_2=0}^{N-1} \frac{h_{k_1}(n,n_0)h_{k_2}(n,m_0)}{1-z(1+s_{k_1})(1+s_{k_2})},$$

   i.e. Eq. (A1)⇔(A8) under valid contour conditions;

2. a 2D two-walker encounter construction under shortcuts where the encounter FPT changes from single-peak to double-peak.

2 A1 → A8: Compact Derivation Roadmap

Let $\alpha_k = 1 + s_k = 1 + q(\cos(2\pi k/N) - 1)$ and

$$\widetilde{P}_{n_0}(n,u) = \sum_{k=0}^{N-1} \frac{h_k(n,n_0)}{1-u\alpha_k}.$$

Insert into Eq. (A1):

$$I(z) = \frac{1}{2\pi i}\oint_C \sum_{k_1,k_2} \frac{h_{k_1}(n,n_0)h_{k_2}(n,m_0)}{(1-u\alpha_{k_1})(u-z\alpha_{k_2})}\,du.$$

For each (k_1,k_2), poles are at $u = 1/\alpha_{k_1}$ (outer family) and $u = z\alpha_{k_2}$ (inner family). Choose a circle $|u| = r$ such that

$$\max_k |z\alpha_k| < r < \min_{k:\alpha_k\neq 0}\left|\frac{1}{\alpha_k}\right|.$$

Then only the inner poles contribute, yielding residues

$$\operatorname*{Res}_{u=z\alpha_{k_2}} = \frac{h_{k_1}(n,n_0)h_{k_2}(n,m_0)}{1-z\alpha_{k_1}\alpha_{k_2}},$$

and summing residues gives Eq. (A8).

3 Numerical Routine (Implementation-Ready)

Parameterize $u = re^{i\theta}$ and use $du/u = i\,d\theta$:

$$I(z) = \frac{1}{2\pi}\int_0^{2\pi} \widetilde{P}_{n_0}(n, re^{i\theta})\widetilde{P}_{m_0}\left(n, \frac{z}{re^{i\theta}}\right)\, d\theta.$$

We evaluate this by trapezoidal rule:

$$I_M = \frac{1}{M}\sum_{j=0}^{M-1} F(re^{2\pi ij/M}).$$

Stability rules used in code.

1. **Contour feasibility check:** enforce $r_{\text{in}} < r < r_{\text{out}}$ with $r_{\text{in}} = \max_k |z\alpha_k|$, $r_{\text{out}} = \min_{\alpha_k \neq 0} |1/\alpha_k|$.
2. **Radius choice:** default $r = \sqrt{r_{\text{in}} r_{\text{out}}}$.
3. **Adaptive refinement:** start from $M = 16$ and double until $|I_{2M} - I_M| < \varepsilon(1 + |I_{2M}|)$.
4. **Cross-check:** compare contour LHS against direct RHS Eq. (A8).

Validation outputs. Table 1 reports four test cases (real/complex z, edge case $q = 0$). Fig. 1 and Fig. 2 show convergence in M and radius invariance.

Table 1: A1/A8 numerical verification summary.

Case	(N, q)	z	M	r	$\lvert LHS - RHS\rvert$	rel.err
R1_real_small	(16,0.30)	$0.40 + 0.00i$	128	0.632	8.13e-18	8.13e-18
R2_complex	(12,0.35)	$0.30 + 0.10i$	64	0.562	1.52e-17	1.52e-17
R3_edge_q0	(20,0.00)	$0.55 + 0.00i$	256	0.742	4.44e-16	1.38e-16
R4_small_z	(18,0.40)	$0.05 + 0.20i$	64	0.454	4.53e-18	4.53e-18

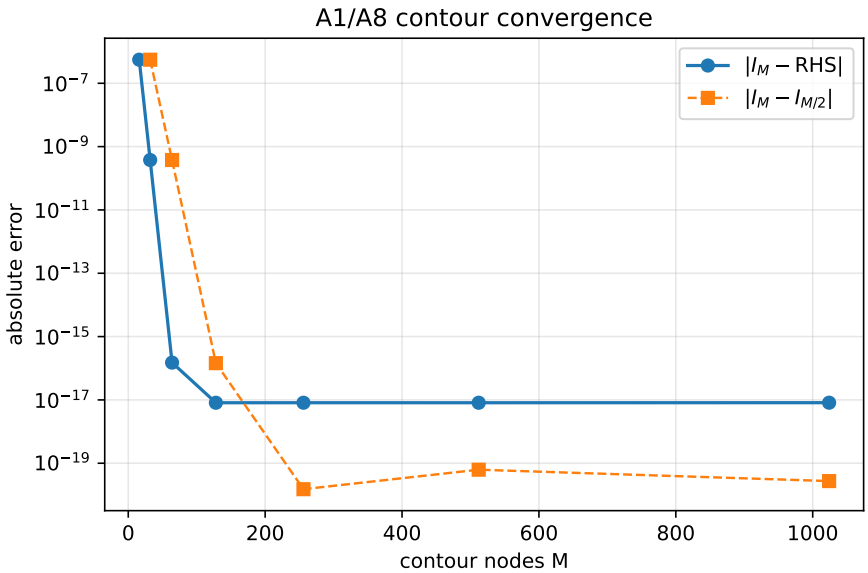

Figure 1: Contour discretization convergence for representative case.

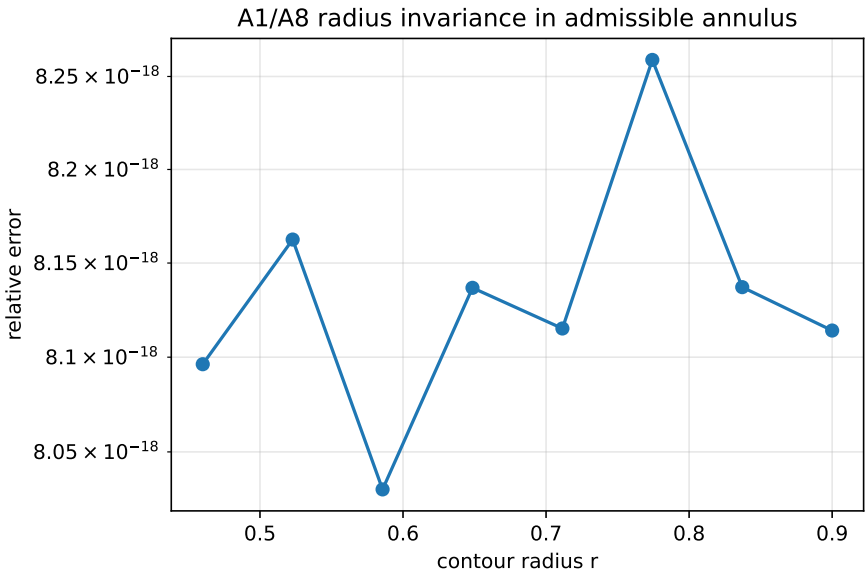

Figure 2: Radius invariance inside admissible annulus.

4 2D Two-Walker Encounter Model With Shortcut

State space is a reflecting $N \times N$ grid for each walker; one-step transitions are lazy biased moves plus an optional directed shortcut from source to destination. We compute first encounter at any site:

$$T_{\text{enc}} = \min\{t \geq 1 : X_t = Y_t\}, \qquad f_{\text{enc}}(t) = \mathbb{P}(T_{\text{enc}} = t).$$

Exact time-domain evolution. Let $J_t(i,j) = \mathbb{P}(X_t = i, Y_t = j, T_{\text{enc}} > t)$. Independent propagation gives

$$J_{t+1} = P_1^\top J_t P_2,$$

encounter flux is diagonal mass

$$f_{\text{enc}}(t+1) = \text{tr}(J_{t+1}),$$

then enforce absorption by zeroing the diagonal of J_{t+1}. This avoids Monte Carlo noise and is fast for moderate grids.

Shortcut channel decomposition. For walker A, we split contributions into trajectories that have used the shortcut vs never used it, by propagating two coupled tensors $(J_t^{(0)}, J_t^{(1)})$. This yields interpretable components "shortcut-used" and "no-shortcut" in $f_{\text{enc}}(t)$.

5 Results: Double-Peak Encounter Under Shortcut

Configuration (representative): $N = 11$, $q = 0.72$, starts $(2,2)$ and $(8,8)$, opposite drifts, walker-A shortcut $(2,2) \to (7,7)$ and weaker walker-B shortcut scaled by 0.1β.

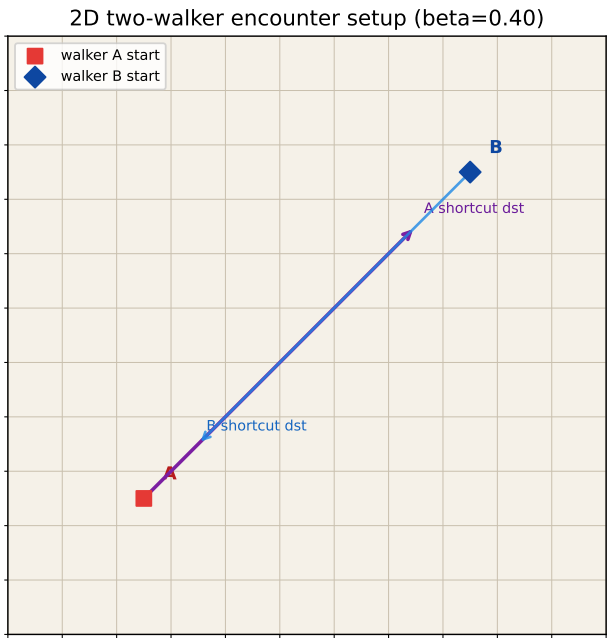

Figure 3: 2D two-walker setup and shortcut geometry.

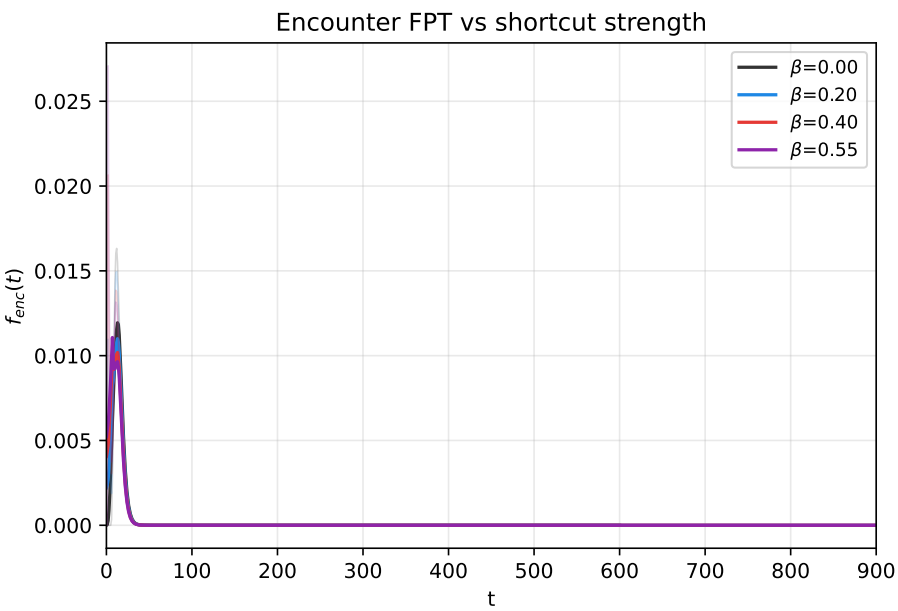

Figure 4: Encounter FPT curves under different shortcut strengths. A transition from single peak ($\beta \lesssim 0.25$) to double peak ($\beta \gtrsim 0.30$) appears.

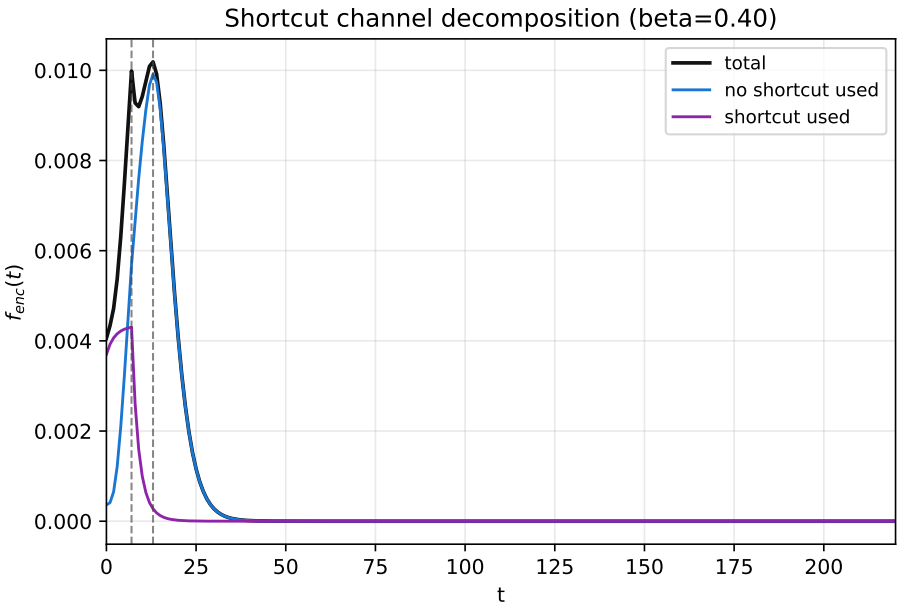

Figure 5: Channel decomposition for representative β: early lobe is shortcut-dominant, later lobe is mostly no-shortcut trajectories.

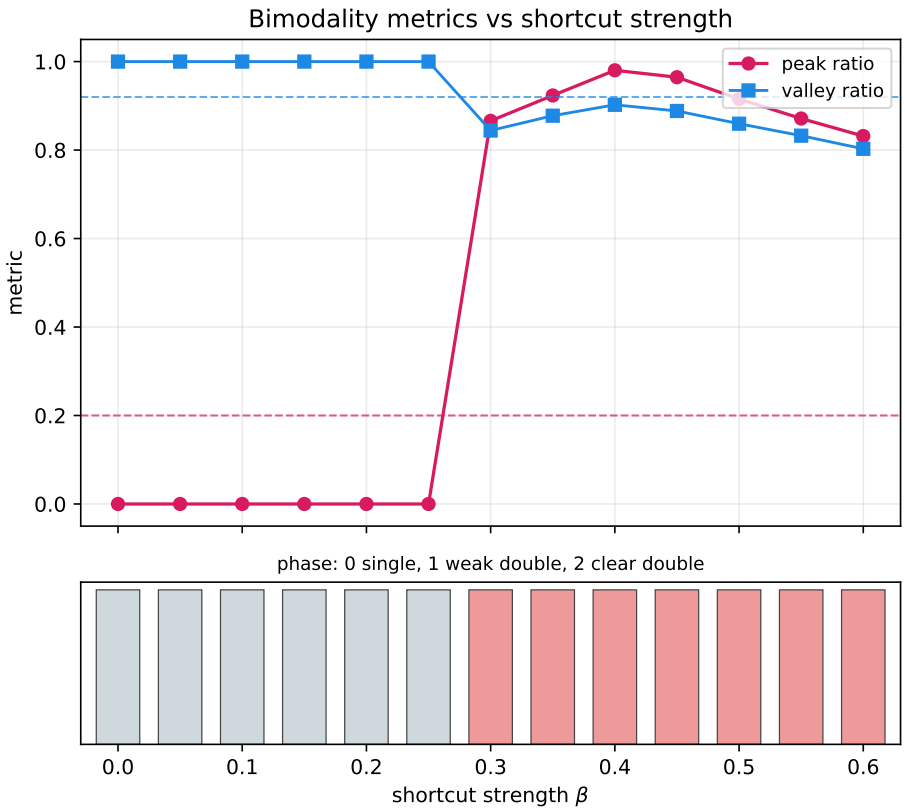

Figure 6: Peak separation together with secondary peak-ratio/valley-ratio diagnostics and phase labels across β.

Table 2: Shortcut scan metrics, including peak separation as the primary phase diagnostic, and phase labels.

β	phase	t_1	t_2	peak ratio	valley ratio	bimodal
0.00	0	-	-	0.000	1.000	0
0.05	0	-	-	0.000	1.000	0
0.10	0	-	-	0.000	1.000	0
0.15	0	-	-	0.000	1.000	0
0.20	0	-	-	0.000	1.000	0
0.25	0	-	-	0.000	1.000	0
0.30	2	7	13	0.866	0.844	1
0.35	2	7	13	0.923	0.878	1
0.40	2	7	13	0.980	0.902	1
0.45	2	7	13	0.965	0.888	1
0.50	2	7	13	0.915	0.860	1
0.55	2	7	13	0.871	0.833	1
0.60	2	7	13	0.832	0.803	1

6 Pitfalls and Debug Checklist

1. **Contour crossing poles (A1/A8):** if $r_{\text{in}} \geq r_{\text{out}}$, do not integrate.
2. **Near-singular denominators:** monitor $\min_{k_1,k_2} |1 - z\alpha_{k_1}\alpha_{k_2}|$.
3. **False micro-peaks in encounter FPT:** we ignore ultra-early transients with $t < t_{\text{ignore}}$ and apply consistent smoothing.
4. **Mass balance:** always track $\sum_t f(t)$ and survival tail at $t_{\max}$.

7 Reproducibility

```
python3 scripts/reportctl.py run --report grid2d_two_walker_encounter_shortcut -- python3 co
python3 scripts/reportctl.py build --report grid2d_two_walker_encounter_shortcut --lang cn
python3 scripts/reportctl.py build --report grid2d_two_walker_encounter_shortcut --lang en
```